

Menulyft HTML & CSS Learning Guide

A complete project-specific guide to the HTML elements, attributes, CSS selectors, properties, layouts, responsive rules and animations used in your restaurant menu website.



HTML tags	CSS properties	CSS classes	JavaScript
31	70	66	0

Project map and learning method

HTML gives the page meaning and structure. CSS controls presentation and responsive behavior. The browser combines both files into the rendered page.

File	Purpose
<code>index.html</code>	Semantic content, sections, cards, links and accessibility.
<code>style.css</code>	Design tokens, spacing, layout, responsive rules and motion.
<code>images/</code>	Local food and About-section photographs.
<code>thali image.jpg</code>	Hero photograph.

How the browser reads the project

```
index.html
-> creates the DOM tree
-> loads style.css through <link>
-> loads local images through 
-> matches CSS selectors to DOM elements
-> calculates layout and paints the page
```

Best study loop

Read one concept, find it in the files, predict a change, edit it, refresh the browser, and explain the result in your own words.

Page structure

```
body
|-- header > nav
|-- main
|   |-- hero
|   |-- today's special
|   |-- menu (4 groups, 8 food cards)
|   |-- offer
|   |-- about
|   |-- reviews
|-- footer#contact
```

The HTML document shell

The document shell sets HTML5 mode, language, encoding, responsive viewport behavior, page metadata and the connection to style.css.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="...">
  <title>Menulyft | Indian Kitchen</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>...</body>
</html>
```

Part	Meaning
<!DOCTYPE html>	Requests modern HTML5 parsing.
lang="en"	Declares English for browsers and assistive technology.
charset="UTF-8"	Supports a broad character set.
viewport	Makes responsive CSS behave correctly on phones.
title	Sets the browser-tab title.
link	Loads the external stylesheet.

DOM idea
Nested HTML creates parent-child relationships. CSS selectors such as .food-image img use those relationships to target descendants.

Every HTML tag used

The current index.html uses 31 different HTML tags. This is the complete extracted inventory.

Tag	Role in this project
<a>	Creates internal links, telephone actions and button-like navigation.
<address>	Marks the footer location information.
<article>	Wraps each independent food card.
<blockquote>	Marks each customer quotation.
<body>	Contains all visible page content.
 	Creates the deliberate line break in the hero heading.
<div>	Neutral wrapper used for component grouping and layout.
	Adds semantic emphasis to part of the hero heading.
<figcaption>	Connects reviewer information to a review figure.
<figure>	Groups a review quotation with its caption.
<footer>	Defines the final Contact Us region.
<h1>	The single primary page heading.
<h2>	Main section headings.
<h3>	Menu category headings.
<h4>	Individual dish names.
<head>	Holds metadata and the stylesheet link.
<header>	Contains the introductory navigation region.
<html>	The root document element.
	Embeds local restaurant and food images.
	One item in the navigation list.
<link>	Loads style.css.
<main>	Wraps the page's unique primary content.
<meta>	Provides encoding, viewport and description information.
<nav>	Marks a group of navigation links.
<p>	Represents paragraph text.
<section>	Defines a themed page region.
<small>	Holds secondary brand text.
	Small generic wrapper for labels and decorative hooks.
	Marks important text such as prices.
<title>	Sets the browser-tab title.
	Unordered list for the navigation items.

Attributes, IDs and accessibility

Attributes add information inside an opening tag. Classes are reusable; IDs should be unique and are used for navigation targets and accessible labelling.

Attribute	Purpose
alt	Text alternative for images.
aria-hidden	Hides decorative content from assistive technology.
aria-label	Provides an accessible name.
aria-labelledby	Connects a region with the heading that labels it.
charset	Declares UTF-8 encoding.
class	Reusable CSS and component hook.
content	Supplies a meta value.
href	Sets a link destination.
id	Creates a unique navigation or labelling target.
lang	Declares the document language.
loading	Lazy-loads off-screen images.
name	Names a metadata field.
rel	Describes a linked resource.
src	Sets the image source path.

Class versus ID

Feature	class	id
Reuse	Many elements may share it.	One unique element.
Selector	.food-card	#menu
Project job	Styling and components.	Navigation and labels.

Internal navigation

```
<a href="#menu">Menu</a>
...
<section id="menu" aria-labelledby="menu-title">
  <h2 id="menu-title">A menu for every mood</h2>
</section>
```

- The skip link jumps directly to main content.
- Every image has alt text.
- Navigation regions have accessible labels.
- Decorative symbols use `aria-hidden="true"`.
- Reduced-motion preferences are respected.

Semantic sections and the food-card component

Semantic elements describe meaning. Reusable class names describe components and their parts.

Region	Key elements/classes	Purpose
Header	header, nav, nav-links	Brand and main navigation.
Hero	section, hero-inner	Primary promise, image and actions.
Menu	section, menu-group, article	Four categories and eight independent dishes.
Offer	section, offer-grid	Weekday promotion.
About	section, about-grid	Story and values.
Reviews	figure, blockquote	Guest quotations and captions.
Footer	footer, address	Ghaziabad, masked phone and copyright.

Food-card anatomy

```
<article class="food-card">
  <div class="food-image">
    
    <span class="food-tag">Bestseller</span>
  </div>
  <div class="food-card-body">
    <div class="food-title-row">
      <h4>Garden Samosa</h4>
      <strong>₹160</strong>
    </div>
    <p>Dish description...</p>
    <a class="order-link" href="tel:...">Order now</a>
  </div>
</article>
```

HTML/CSS limitation

Without JavaScript, an order link can navigate or start a telephone action, but it cannot maintain cart state or quantities.

CSS syntax, selectors and the cascade

A CSS rule contains a selector and a declaration block. The selector finds elements; each property-value declaration changes one aspect of their presentation.

```
.food-card {  
  overflow: hidden;  
  border: 1px solid var(--line);  
  background: var(--paper);  
}  
.food-card:hover {  
  box-shadow: var(--shadow);  
  transform: translateY(-7px);  
}
```

Selector type	Project example	Meaning
Universal	*	Every element.
Type	body, img, a	All tags with that name.
Class	.food-card	Elements carrying that class.
Descendant	.food-image img	Images nested inside .food-image.
Child	.section-heading > p	Direct paragraph child only.
Pseudo-class	:hover, :focus	Interactive or structural state.
Pseudo-element	::before, ::after	Generated decorative box.
Negation	:not(.eyebrow)	Items without a condition.

Cascade and specificity

- Type selectors are weaker than classes.
- More specific selectors beat less specific selectors.
- When specificity is equal, the later rule wins.
- Media-query rules work because they appear later and target the same classes.
- Inherited properties such as color and font-family flow from parents to children.

Design tokens, units and functions

The `:root` block stores reusable colors, shadows and font stacks as custom properties. `var()` reads those values throughout the stylesheet.

```
:root {  
  --ink: #1e241e;  
  --forest: #23372b;  
  --cream: #f5efe3;  
  --saffron: #d9822b;  
  --line: rgba(30, 36, 30, 0.16);  
  --shadow: 0 20px 60px rgba(46, 42, 31, 0.12);  
}
```

Unit/function	Meaning
px	Reference pixels for borders, gaps and breakpoints.
rem	Text size relative to the root font size.
%	Value relative to a containing dimension.
vw	One percent of viewport width.
fr	Fraction of available Grid space.
var()	Reads a custom property.
calc()	Combines different values.
min()	Selects the smaller value.
clamp()	Fluid value with minimum and maximum limits.
rgba()	Color with alpha transparency.
repeat()/minmax()	Concise flexible Grid definitions.
linear-gradient()	Generated color/image transition.
translateY()/scale()/rotate()	Transforms an element visually.

Box model, spacing and sizing

Every element is a box made from content, padding, border and margin. The universal reset makes dimensions predictable.

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

.container {
  width: min(1180px, calc(100% - 48px));
  margin-inline: auto;
}

.section { padding: 108px 0; }
```

Layer	Meaning
Content	Text, image or child elements.
Padding	Space between content and border.
Border	Visible edge around padding/content.
Margin	Space outside the element.
gap	Space between Flexbox or Grid children.

Why border-box

A declared width includes padding and border, preventing boxes from unexpectedly becoming wider than their container.

Flexbox and CSS Grid

Flexbox arranges items mainly along one axis. Grid defines rows and columns together. Menulyft uses both, often nested inside each other.

Flexbox navigation

```
.nav {
  display: flex;
  align-items: center;
  min-height: 88px;
  gap: 42px;
}
.brand { margin-right: auto; }
```

Grid food cards

```
.menu-grid {
  display: grid;
  grid-template-columns: repeat(2, minmax(0, 1fr));
  gap: 26px;
}
```

Tool	Use when	Important properties
Flexbox	Items move mainly in a row or column.	flex-direction, align-items, justify-content, flex-wrap, gap
Grid	Rows and columns must be coordinated.	grid-template-columns, grid-column, gap, justify-self
Both	A component contains smaller aligned groups.	Grid outside, Flexbox inside is common.

Positioning, imagery and visual treatment

Positioning creates overlays and sticky labels. Image rules create consistent crops while borders, shadows, gradients and filters add polish.

```
.hero-visual { position: relative; }
.hero-image-wrap {
  position: absolute;
  inset: 0;
  overflow: hidden;
}
.floating-note {
  position: absolute;
  right: -28px;
  bottom: 42px;
  z-index: 2;
}
```

position value	Behavior
relative	Keeps normal flow and creates a positioning reference.
absolute	Removes the box from normal flow and uses offsets.
fixed	Pins an element to the viewport.
sticky	Acts normal until a scroll threshold, then sticks.
static	Default normal-flow positioning.

Image cropping

```
.food-image { height: 245px; overflow: hidden; }
.food-image img {
  width: 100%;
  height: 100%;
  object-fit: cover;
}
```

Common bug

If an absolutely positioned child jumps to the page edge, confirm that its intended parent has position: relative.

Typography, hover effects and animation

The project combines a serif display face with a sans-serif interface face. Hover and focus states provide feedback, while keyframes animate the hero entrance.

Typography role	Treatment
Display headings	Serif, large size, tight line-height, negative letter-spacing.
Body/interface	Sans-serif, readable base size and moderate line-height.
Labels/eyebrows	Small uppercase sans-serif with wide letter-spacing.
Fluid headings	clamp(minimum, viewport-based value, maximum).

Transition

```
.food-card {  
  transition: transform 0.35s ease, box-shadow 0.35s ease;  
}  
.food-card:hover {  
  box-shadow: var(--shadow);  
  transform: translateY(-7px);  
}
```

Keyframes

```
@keyframes reveal-up {  
  from { opacity: 0; transform: translateY(24px); }  
  to   { opacity: 1; transform: translateY(0); }  
}  
.hero-copy { animation: reveal-up 0.8s ease both; }
```

Reduced motion

```
@media (prefers-reduced-motion: reduce) {  
  *, *::before, *::after {  
    animation-duration: 0.01ms !important;  
    transition-duration: 0.01ms !important;  
  }  
}
```

Responsive design and media queries

The desktop design is the default. Later media-query rules override only what must change as available width decreases.

Breakpoint	Main changes
1000px	Tighter navigation, hero proportions, menu labels and image heights.
760px	Navigation links hide; main grids stack; footer becomes two columns.
480px	Smaller gutters; action buttons, special card and footer stack.
Reduced motion	Smooth motion is minimized for accessibility.

```
@media (max-width: 760px) {  
  .menu-group { grid-template-columns: 1fr; }  
  .menu-grid { grid-template-columns: 1fr; }  
  .group-title { position: static; }  
}
```

- Use fluid containers instead of fixed page width.
- Stack columns when content becomes cramped.
- Keep touch targets large through padding and min-height.
- Allow the category strip to scroll horizontally rather than crush labels.
- Test content at desktop, tablet and phone widths.

Complete CSS property glossary

This table is generated from the current style.css and contains all 70 declared standard properties with a real example value.

Property	Meaning	Example
align-items	Aligns all flex/grid children on the cross axis.	center
align-self	Overrides alignment for one item.	stretch
animation	Runs a named keyframe sequence.	reveal-up 0.8s ease both
animation-duration	Sets animation time.	0.01ms !important
animation-iteration-count	Sets repetition count.	1 !important
background	Sets color, image or gradient background.	var(--paper)
border	Sets border width, style and color.	1px solid transparent
border-bottom	Sets only the bottom border.	1px solid rgba(255, 255, 255, 0.12)
border-color	Changes border color.	var(--saffron)
border-left	Sets only the left border.	1px solid rgba(255, 255, 255, 0.25)
border-radius	Rounds corners.	50% 50% 45% 55%
border-top	Sets only the top border.	1px solid var(--line)
bottom	Offsets a positioned element from the bottom.	4px
box-shadow	Draws a shadow around a box.	0 32px 80px rgba(0, 0, 0, 0.3)
box-sizing	Controls whether width includes padding and border.	border-box
color	Sets foreground text color.	var(--ink)
content	Creates pseudo-element content.	""
display	Chooses block, flex, grid or hidden behavior.	block
filter	Applies image effects such as saturation.	saturate(0.88)
flex	Controls flex growth, shrinkage and basis.	1
flex-direction	Sets row or column direction.	column
flex-wrap	Allows items to wrap onto more lines.	wrap
font-family	Sets typeface and fallbacks.	var(--sans)
font-size	Sets text size.	16px
font-style	Sets normal or italic style.	normal
font-weight	Sets text thickness.	700
gap	Spaces grid/flex items.	42px
grid-column	Places an item between grid lines.	2 / -1
grid-template-columns	Defines grid columns.	1.05fr 0.95fr
height	Sets box height.	41px
inset	Shorthand for all four position offsets.	0
justify-content	Distributes items on the main axis.	center
justify-self	Aligns one grid item in its area.	start

Property	Meaning	Example
left	Offsets a positioned element from the left.	8px
letter-spacing	Changes character spacing.	0.16em
line-height	Controls text line spacing.	1.65
list-style	Controls or removes list markers.	none
margin	Sets outer spacing.	0
margin-bottom	Sets outer spacing below.	44px
margin-inline	Sets logical left/right margins.	auto
margin-right	Sets outer spacing on the right.	auto
margin-top	Sets outer spacing above.	5px
max-width	Caps width.	100%
min-height	Sets a minimum height.	50px
min-width	Sets a minimum width.	max-content
object-fit	Controls image fitting inside a fixed box.	cover
opacity	Sets transparency.	0
overflow	Controls content outside a box.	hidden
overflow-x	Controls horizontal overflow.	auto
padding	Sets inner spacing.	0
padding-bottom	Sets inner spacing below.	72px
padding-left	Sets inner spacing on the left.	0
padding-top	Sets inner spacing above.	150px
place-items	Aligns grid items in both axes.	center
position	Selects static, relative, absolute, fixed or sticky positioning.	fixed
right	Offsets a positioned element from the right.	0
row-gap	Sets row spacing.	8px
scroll-behavior	Controls smooth or instant scrolling.	smooth
scroll-padding-top	Reserves space above anchor targets.	100px
text-align	Aligns inline content.	right
text-decoration	Adds or removes underline and similar decoration.	none
text-transform	Visually changes text casing.	uppercase
top	Offsets a positioned element from the top.	8px
transform	Moves, scales or rotates an element.	translateY(-150%)
transform-origin	Sets the transform pivot.	right
transition	Animates a property change.	transform 0.2s ease
transition-duration	Sets transition time.	0.01ms !important
white-space	Controls wrapping and spaces.	nowrap
width	Sets box width.	min(1180px, calc(100% - 48px))
z-index	Controls stacking order.	1000

Class, ID and token inventory

These names form the project's component vocabulary. Classes are reusable styling hooks; IDs are unique destinations and labelling targets.

All IDs

about, about-title, beverages, contact, desserts, home, main-content, mains, menu, menu-title, offer-title, reviews-title, special-title, starters

All classes

Group	Classes
1	about-copy, about-grid, about-image, about-section, about-values, brand, brand-mark, button, button-dark, button-light, button-primary
2	category-list, container, eyebrow, featured-review, floating-note, food-card, food-card-body, food-image, food-tag, food-title-row, footer-bottom
3	footer-brand, footer-column, footer-intro, footer-phone, footer-top, group-title, hero, hero-actions, hero-copy, hero-image-wrap, hero-inner
4	hero-notes, hero-text, hero-visual, lead, menu-grid, menu-group, menu-section, nav, nav-cta, nav-links, offer-grid
5	offer-label, offer-main, offer-section, offer-side, order-link, review-card, review-heading, reviews-grid, reviews-section, section, section-heading
6	site-footer, site-header, skip-link, special-card, special-copy, special-number, special-price, special-section, stars, text-link, veg-dot

Custom properties

Token	Stored value
--ink	#1e241e
--forest	#23372b
--forest-light	#304b39
--cream	#f5efe3
--paper	#fffaf0
--saffron	#d9822b
--rust	#aa4b2b
--line	rgba(30, 36, 30, 0.16)
--shadow	0 20px 60px rgba(46, 42, 31, 0.12)
--serif	"DM Serif Display", Georgia, serif
--sans	"DM Sans", Arial, sans-serif

Naming lesson

Choose names based on role, not temporary appearance. `.food-card` remains useful after a redesign; `.white-box` may not.

Practice plan and debugging

Learning becomes durable when you rebuild components and explain why each rule exists.

Level 1 - Visual edits

- Change `--saffron` and `--rust` to make a new palette.
- Adjust section padding and observe page rhythm.
- Change heading clamp limits and test three widths.
- Compare subtle and strong food-card shadows.
- Change image object-position and observe the crop.

Level 2 - Structural edits

- Duplicate a food-card and replace all content.
- Add a fifth category and matching menu group.
- Create three desktop card columns and choose a new breakpoint.
- Add a new footer column using `footer-column`.
- Build a second review-card variant with one extra class.

Debugging checklist

- Is `style.css` linked correctly?
- Does the class name match exactly?
- Is a later rule or media query overriding the value?
- Is the element really a Grid/Flex item?
- Does the absolute child have a relative parent?
- Is overflow clipping it?
- Does every image path exist?
- Does each href hash match an ID?
- Are every tag and CSS brace closed?
- Did you test desktop, tablet, phone, keyboard and reduced motion?

Real front-end skill

Do not try to memorize every property. Learn to recognize the layout problem, locate the responsible component, choose a suitable CSS tool and verify the result at multiple widths.

QUICK REFERENCE

Project facts at a glance

Item	Current project
Technology	HTML5 and CSS3 only
JavaScript	None
Layouts	Flexbox and CSS Grid
Breakpoints	1000px, 760px and 480px
Menu	4 categories and 8 food cards
Footer	Ghaziabad and 9050XXXX
Files	index.html and style.css

Essential patterns

```
/* Center content */
.container { width: min(1180px, calc(100% - 48px)); margin-inline: auto; }

/* Two equal columns */
.menu-grid { display: grid; grid-template-columns: repeat(2, minmax(0, 1fr)); }

/* Stack for mobile */
@media (max-width: 760px) { .menu-grid { grid-template-columns: 1fr; } }

/* Smooth hover feedback */
.food-card { transition: transform 0.35s ease; }
.food-card:hover { transform: translateY(-7px); }
```

Study. Change. Predict. Refresh. Explain. Repeat.